

CS598: Cloud Computing Capstone

Project Task 2: Stream Processing Big Data in AWS

By: Chris Johnson

University of Illinois at Urbana-Champaign

July 2020

Task 2- Project Vision:

*Use Spark Streaming in AWS to Efficiently Query a
Massive Airline Data Set (20+ GB)*



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Task 2: Overview & System Design

Key Highlighted Areas: System Architecture, Batching vs. Streaming Systems, and System-Level Design Optimizations

Summary

In Task 2, I used Kafka & Spark Streaming to process streaming airline data then save in AWS as necessary. The specific tools used were AWS Managed Streaming for Kafka (MSK) and Spark Structured Streaming with Pyspark in AWS Glue. This architecture was heavily influenced by two of AWS's latest articles on stream processing, which were published in April 2020 and are available below:

- <https://aws.amazon.com/about-aws/whats-new/2020/04/aws-glue-now-supports-serverless-streaming-etl/>
- <https://aws.amazon.com/blogs/aws/new-serverless-streaming-etl-with-aws-glue/>

Comparison of Streaming vs. Batch Processing

Personally, I found the infrastructure setup for stream processing was much more complex than for batch processing. Instead of simply loading data from S3, stream processing involved the creation of a Kafka cluster (which was complex), then linking that to Spark Streaming tools (which was also highly involved). Despite the complexity, it is evident that the future is moving towards streaming tools and new market offerings from major cloud players are quickly evolving. After experiencing both systems, I would recommend a batch processing system because of the bulk of data and the fact that the flight data is historical (i.e. from 20 years of flight data vs. live data coming in).

Key Step #1: Set up Kafka Cluster

Set up a VPC in AWS, then specify and launch Kafka cluster.

The cluster used in this case had 2 availability zones, with 2 brokers each for fault-tolerance & reliability. A schematic of AWS MSK is available below, along with detail on my cluster configuration.

Architecture Optimization: AWS Managed Streaming for Kafka (MSK): AWS MSK runs open-source versions of Kafka and manages the provisioning and maintenance of resources in MSK clusters. This frees users from the task of managing the operational overhead of the Kafka environment and enables more time to be allocated towards application development.



General

Cluster ARN

arn:aws:kafka:us-east-1:069691451703:cluster/AWSKafkaTutorialCluster/86db14c3-6ddd-407c-b846-9d7825f76a38-1

Status

Active

Creation time

July 10, 2020, 9:26:36 PM EDT

Apache Kafka version

2.2.1

Networking

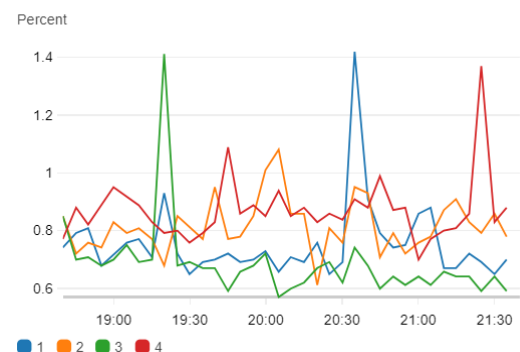
Subnets (Availability Zones)

subnet-06bdd043023413bee
subnet-040cfb7179925f0af

Security groups applied

sg-0d8a0fb173c94ae15

CPU (User) usage by broker



Key Step #2: Create a Kafka Topic

Create a topic in Kafka, build a producer to populate the topic, then build a test consumer

My next major step was to stream the cleaned data in S3 from Task 1 to the Kafka cluster. To do so, I launched an EC2 instance and connected it with the Kafka cluster. Then, I created a topic and ran the producer (more detail discussed in the video demo).

Key Step #3: Process & Transform Data with Spark Structured Streaming, then Save As Needed

There are 3 major steps for a Spark Streaming & Glue ETL job, with my example shown below:

1. Create the AWS Glue connection to the Amazon MSK cluster.
2. Set the Data Catalog table for the streaming Kafka topic.
3. Write the Pyspark script for the Spark Streaming processing to get the desired outputs.

Architecture Optimization: Spark & AWS Glue: AWS Glue is a managed Extract, Transform, Load service which uses Spark Structured Streaming and Pyspark. AWS Glue is 'serverless', which means AWS handles the infrastructure portion- this gives me more time to focus on data analytics. It also can be more cost effective, as I basically only pay for what I use when I use it.

The screenshot displays the AWS Glue console interface. On the left, a code editor shows a Pyspark script for a Spark Streaming job. The script includes imports for SparkContext, GlueContext, and DynamicFrame, and defines a job named 'cs598streamjob-production'. The script uses the 'flighttimingschema' table from the 'flighttimingschema' database and writes the output to the 'cs598streamingschema' table. On the right, the 'IAM role' configuration is shown, set to 'glueRole-kafkaSparkStream'. Below the IAM role, the 'Type' is set to 'Spark Streaming' and the 'Glue version' is set to 'Spark 2.4, Python 2 (Glue Version 1.0)'. A table at the bottom lists the jobs, with 'cs598streamjob-production' selected. The table has columns for Name, Type, and ETL. The script editor at the bottom shows the following code:

```
1 from pyspark.sql import DataFrame, Row
2 import datetime
3 from aws glue import DynamicFrame
4
5 ## @params: [JOB_NAME]
6 args = getResolvedOptions(sys.argv, ['JOB_NAME'])
7
8 sc = SparkContext()
9 glueContext = GlueContext(sc)
10 spark = glueContext.spark_session
11 job = Job(glueContext)
12 job.init(args['JOB_NAME'], args)
13
14 ## @type: DataSource
15 ## @args: [database = "flighttimingschema", table_name = "cs598streamingschema", additionalOptions = {"startingOffsets": "earliest", "inferSchema": "false"}, stream_t
16 ## @return: DataSource
17
18 ## @inputs: []
19
20 datasource0 = glueContext.create_data_frame.from_catalog(database = "flighttimingschema", table_name = "cs598streamingschema", transformation_ctx = "datasource0", add
21 ## @type: DataSink
22
23 ## @args: [path = "s3://aws-glue-.../cs598streamingschema", format = "parquet", partition_cols = ["date"], write_mode = "overwrite", compression = "snappy", row_group_size_in_bytes = 1048576, row_
```

Task 2: Group 1 Questions

Group 1: Do My Results Make Sense & Are They Useful? Yes, the results make sense based on my experience with air travel, as I fly 50,000 to 100,000 miles per year for work. They are also in-line with the answers given by the professor.

- 1.1: Chicago (ORD), Atlanta (ATL), Dallas Fort-Worth (DFW), and Los Angeles (LAX) are by far the most popular airports as they are hubs. This gives users a good sense of where they will be connecting.
- 1.2: The average on-time arrival performance for the top 10 airlines was between 0 and 10 minutes, which is in-line with my expectations as well as the given answers. Note- I used DepDelayMinutes instead of DepDelay, because I did not want to include negative values in my calculation of on-time performance (i.e. early = on-time).

Group 1: Approach & Algorithms For both Group 1 questions, I used Spark Structured Streaming with Pyspark to obtain only the fields that I wanted from the Kafka topic and map the fields to the correct data types. Then, I took the top 10 for each question. For Question 1.1, I had to use 'JOIN'.

<i>Question 1.1 Results: Rank the top 10 most popular airports by numbers of flights to/from the airport.</i>			<i>Question 1.2 Results: Rank the top 10 airlines by on-time arrival performance (shown in minutes).</i>		
1	ORD	12449353	1	HA	3.9723442136498517
2	ATL	11540422	2	AQ	5.044499594078162
3	DFW	10799303	3	PS	5.672566797335321
4	LAX	7723595	4	ML (1)	8.703598142334235
5	PHX	6585534	5	WN	9.128027655975494
6	DEN	6273787	6	F9	9.933341238438784
7	DTW	5636622	7	PA (1)	10.344808759846858
8	IAH	5480734	8	US	10.531429711732809
9	MSP	5199213	9	NW	10.584071568049064
10	SFO	5171033	10	OO	10.660752365340201

Task 2: Group 2 Questions

Group 2: Do My Results Make Sense & Are They Useful? Yes, as they allow users to choose the best performing airlines for several popular airports in the US. They also show average delays for several key source-destination pairs. The results are largely in-line with the answers given by the professor as well. Notes:

- 2.1: There was a tight band of departure performance across the stated airports (i.e. 0 to 14 minutes). I used DepDelayMinutes instead of DepDelay, because I did not want to negative values to impact my calculations.
- 2.3 & 2.4: I used ArrDelayMinutes instead of ArrDelay, because I did not want negative values to impact results.

Group 2: Approach & Algorithms For each of the Group 2 questions, I used Spark Structured Streaming with Pyspark to obtain only the fields that I wanted from the Kafka topic, map the correct data types, filter the airports I needed, and save as necessary. Then, I took the top 10 results for each airport.

Question 2.1 Results: For each airport X, rank the top-10 carriers in decreasing order of on-time <u>departure</u> performance from X. SRQ <table><tr><td>AA</td><td>4.903171456888008</td></tr><tr><td>UA</td><td>5.156787092248334</td></tr><tr><td>US</td><td>5.8389443161153975</td></tr><tr><td>TZ</td><td>5.931164901664145</td></tr><tr><td>TW</td><td>6.207765195285417</td></tr><tr><td>DL</td><td>6.281154243819908</td></tr><tr><td>XE</td><td>6.568776772965255</td></tr><tr><td>EA</td><td>7.0678502373000525</td></tr><tr><td>YV</td><td>7.180073126142596</td></tr><tr><td>NW</td><td>7.303365563406471</td></tr></table>	AA	4.903171456888008	UA	5.156787092248334	US	5.8389443161153975	TZ	5.931164901664145	TW	6.207765195285417	DL	6.281154243819908	XE	6.568776772965255	EA	7.0678502373000525	YV	7.180073126142596	NW	7.303365563406471	JFK <table><tr><td>UA</td><td>7.65507045797685</td></tr><tr><td>CO</td><td>10.82607790043741</td></tr><tr><td>AA</td><td>11.276934678103498</td></tr><tr><td>PA (1)</td><td>12.054336966394187</td></tr><tr><td>XE</td><td>12.077472527472528</td></tr><tr><td>DH</td><td>12.146237766725493</td></tr><tr><td>DL</td><td>13.2652202872705</td></tr><tr><td>B6</td><td>13.382045750569848</td></tr><tr><td>TW</td><td>13.63391443023823</td></tr><tr><td>US</td><td>14.01086774786825</td></tr></table> CMH <table><tr><td>ML (1)</td><td>4.370341614906832</td></tr><tr><td>AA</td><td>5.303880929871767</td></tr><tr><td>PI</td><td>5.79635079458505</td></tr><tr><td>EA</td><td>5.952433628318584</td></tr><tr><td>NW</td><td>6.184962176464602</td></tr><tr><td>DL</td><td>6.264886380276628</td></tr><tr><td>TW</td><td>6.955588049773496</td></tr><tr><td>US</td><td>7.2963704060180845</td></tr><tr><td>DH</td><td>7.7534329563812605</td></tr><tr><td>WN</td><td>8.536949177348733</td></tr></table>		UA	7.65507045797685	CO	10.82607790043741	AA	11.276934678103498	PA (1)	12.054336966394187	XE	12.077472527472528	DH	12.146237766725493	DL	13.2652202872705	B6	13.382045750569848	TW	13.63391443023823	US	14.01086774786825	ML (1)	4.370341614906832	AA	5.303880929871767	PI	5.79635079458505	EA	5.952433628318584	NW	6.184962176464602	DL	6.264886380276628	TW	6.955588049773496	US	7.2963704060180845	DH	7.7534329563812605	WN	8.536949177348733	BOS <table><tr><td>PA (1)</td><td>4.760287290186898</td></tr><tr><td>ML (1)</td><td>5.760416666666667</td></tr><tr><td>TZ</td><td>7.452451269935026</td></tr><tr><td>DL</td><td>8.422345377132867</td></tr><tr><td>EV</td><td>8.538341158059469</td></tr><tr><td>EA</td><td>9.017153017466793</td></tr><tr><td>NW</td><td>9.27271332212635</td></tr><tr><td>US</td><td>10.199998308360506</td></tr><tr><td>AA</td><td>10.283630290413734</td></tr><tr><td>TW</td><td>11.109131029189552</td></tr></table> SEA <table><tr><td>PS</td><td>5.006949270326616</td></tr><tr><td>OO</td><td>5.109358345768493</td></tr><tr><td>DL</td><td>7.530515296761439</td></tr><tr><td>YV</td><td>7.6989051094890515</td></tr><tr><td>US</td><td>7.844094309312477</td></tr><tr><td>NW</td><td>8.188894389300488</td></tr><tr><td>AA</td><td>8.300998183659363</td></tr><tr><td>WN</td><td>8.738621761173805</td></tr><tr><td>CO</td><td>8.967416423844861</td></tr><tr><td>EV</td><td>9.251851851851852</td></tr></table>		PA (1)	4.760287290186898	ML (1)	5.760416666666667	TZ	7.452451269935026	DL	8.422345377132867	EV	8.538341158059469	EA	9.017153017466793	NW	9.27271332212635	US	10.199998308360506	AA	10.283630290413734	TW	11.109131029189552	PS	5.006949270326616	OO	5.109358345768493	DL	7.530515296761439	YV	7.6989051094890515	US	7.844094309312477	NW	8.188894389300488	AA	8.300998183659363	WN	8.738621761173805	CO	8.967416423844861	EV	9.251851851851852
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Question 2.4: For each source-dest. pair X-Y, determine the mean arr. delay (in min.) for a flight from X to Y. LGA to BOS AverageDelayMin ▾ 1 7.655719892412466	BOS to LGA AverageDelayMin ▾ 1 8.500401514282437 OKC to DFW AverageDelayMin ▾ 1 8.376757753377914	MSP to ATL AverageDelayMin ▾ 1 12.067839513985572

Task 2: Group 3 Questions

Question 3.2: Find for each X-Y-Z and day/month (dd/mm) combination in the year 2008, the two flights (X-Y and Y-Z) that satisfy constraints (a) and (b) and have the best individual performance with respect to constraint (c), if such flights exist. Constraints:

- The second leg of the journey (flight Y-Z) must depart two days after the first leg (flight X-Y). For example, if X-Y departs on January 5, 2008, Y-Z must depart on January 7, 2008.
- Tom wants his flights scheduled to depart airport X before 12:00 PM local time and to depart airport Y after 12:00 PM local time.
- Tom wants to arrive at each destination with as little delay as possible. Assume you know the delay of each flight.

Q3.2 Do My Results Make Sense & Are They Useful? Yes- the results make sense and matched exactly with the answers provided by the instructors. However, given the specificity of the flight constraints the results are not broadly useful.

Q3.2 Approach & Algorithms: Due to the size of the data set and CPU-intensive nature of the queries, this question required a slightly different approach from the others in Groups 1 & 2. To make these queries more efficient from a run-time and resource perspective, I used multiple stages. First, I used Pyspark to select the fields I needed, map the data types correctly, then filter out the origins/dates needed. Then, I created a joined table of the potential flight combinations which was used to perform the best flight analysis.

Q3.2 Results

BOS → ATL → LAX, 03/04/2008

FirstAirline ▾	FirstFlightNum ▾	SecondAirline ▾	SecondFlightNum ▾	TotalArrivalDelay ▾
FL	270	FL	40	5.0

PHX → JFK → MS, 07/09/2008

FirstAirline ▾	FirstFlightNum ▾	SecondAirline ▾	SecondFlightNum ▾	TotalArrivalDelay ▾
B6	178	NW	609	-42.0

DFW → STL → ORD, 24/01/2008

FirstAirline ▾	FirstFlightNum ▾	SecondAirline ▾	SecondFlightNum ▾	TotalArrivalDelay ▾
1 AA	1336	AA	2245	-19.0

LAX → MIA → LAX, 16/05/2008

FirstAirline ▾	FirstFlightNum ▾	SecondAirline ▾	SecondFlightNum ▾	TotalArrivalDelay ▾
AA	280	AA	456	-9.0